1.Content

Part 1. Number Systems and Proof Techniques

Part 2. Set Theory

Part 3. Relations

Part 4. Functions

Part 5. Propositional Logic and First-Order Predicate Logic

Part 6. Combinatorics, Discrete probability and Statistics

2.Proof Technique

Finding a counter-example 找反例

Proof by contradiction 反证法

ex: √ 2 is not a rational number

If √ 2 were rational then we could write it as √ 2 = x/y where

x and y are integers and y is not 0.

By repeatedly cancelling common factors（公因数）, we can make sure

that x and y have no common factors so they are not both even.

Then 2 = x 2 / y 2 so x 2 = 2y 2 so x 2 is even. This means x is

even, because the square of any odd number is odd

Let x = 2w for some integer w.

Then x 2 = 4w 2 so 4w 2 = 2y 2 so y 2 = 2w 2 so y 2 is even so y

is even. This contradicts the fact that x and y are not both even, so our

original assumption, that √ 2 is rational, must have been wrong

Proof by induction

Base Case:

Show that the property holds for $n = 0$.

Inductive Step:

Assume that the property holds for $n = m$.

Show that it holds for $n = m + 1$.

Conclusion:

You can now conclude that the property holds for

every natural number n.

3.Basic Datatype

3.1Natural Numbers

$N = \{0, 1, 2, 3, \ldots\}$

We say that N is closed under addition 封闭式加法 because if you take any two numbers in N and add them, you always get another number in N.

Key property: Any natural number can be obtained from 0 by applying the operation $S(n) = n + 1$ some number times.

Examples: $S(0) = 1$. $S(S(0)) = 2$. $S(S(S(0))) = 3$.

3.2Integer

$Z = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

The positive integers: $Z_+ = \{1, 2, 3, \ldots\}$.

3.3Rational Numbers

Q is the set of all numbers that can be written as x/y where x and

y are integers and y is not 0.

3.4Real Numbers

R is the set of all (decimal) numbers — distances to points on a

number line. A real number that is not rational is called irrational.

3.5Prime Number

A prime number is a integer greater than 1 which has exactly two divisors that are

positive integers: 1 and itself.

Every integer greater than 1 can be written as a unique product of prime numbers.

4. Set Theory

4.1 Set

A *set* is a collection of objects, called the *elements* of the set.

We have written down the elements of each set and contained them between the

braces { }.

For a large set, especially an infinite set, we cannot write down all

the elements. We use a predicate P instead. $S = \{x \mid P(x)\}$

denotes the set of objects x for which the predicate $P(x)$ is true.

The empty set has no elements. It is written as $\varnothing$ or as {}.

4.2 Subsets

A set B is called a *subset* of a set A if every element of B is an element of A. This is denoted by $B \subseteq A$.

$\{3, 4, 5\} \subseteq \{1, 5, 4, 2, 1, 3\}$, $\{3, 3, 5\} \subseteq \{3, 5\}$, $\{5, 3\} \subseteq \{3, 5\}$

4.3Equality

A set A is called *equal* to a set B if $A \subseteq B$ and $B \subseteq A$. This is denoted by $A = B$.

$\{1\} = \{1, 1, 1\}$, $\{1, 2\} = \{2, 1\}$,

$\{5, 4, 4, 3, 5\} = \{3, 4, 5\}$.

4.4Data types

Modern computing languages require that variables are declared

as belonging to a particular *data type*; a data type consists of

a set of objects; a list of standard operations on those objects

Specifying the type of a variable is equivalent to specifying the set

*from which the variable is drawn.*当你写下 `int x;` 时，你其实是在告诉计算机：

“变量 x 只能从 **整数集（Integer Set）** 中取值”

4.5UNION of two sets

Definition The union of two sets A and B is the set

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.

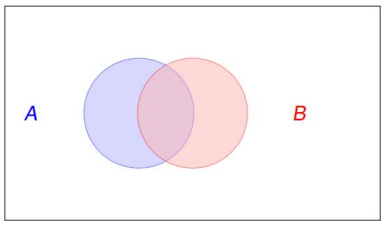


Figure: Venn diagram of $A \cup B$.

4.6 Intersection of two sets

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

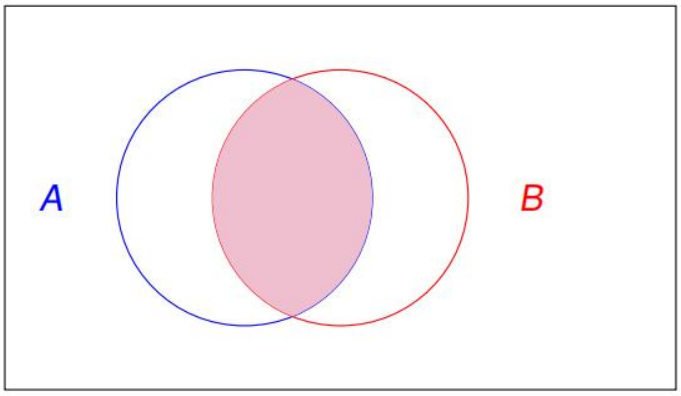


Figure: Venn diagram of $A \cap B$.

4.7 Relative Complement

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}.$$

ex:

$A = \{4, 7, 8\}$

$B = \{4, 9, 10\}$.

$A - B = \{7, 8\}$

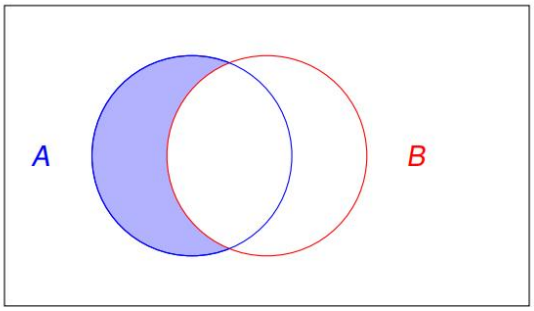


Figure: Venn diagram of $A - B$.

4.8Complement: When we are dealing with subsets of some large set

U, then we call U the *universal set* for the problem in question.

The complement of a set A is the set

$$\sim A = \{x \mid x \notin A\} = U - A.$$

4.9 Symmetric

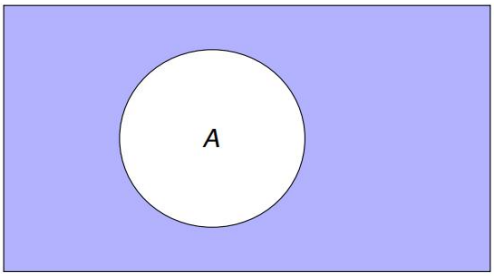


Figure: Venn diagram of $\sim A$.

difference

The symmetric difference of two sets A and B is the set

$A \Delta B = \{x \mid (x \in A \text{ and } x < B) \text{ or } (x < A \text{ and } x \in B)\}$.

$A = \{4, 7, 8\}$ and $B = \{4, 9, 10\}$.

$A \Delta B = \{7, 8, 9, 10\}$

Definition The symmetric difference of two sets A and B is the set

$$A \Delta B = \{x \mid (x \in A \text{ and } x \notin B) \text{ or } (x \notin A \text{ and } x \in B)\}.$$

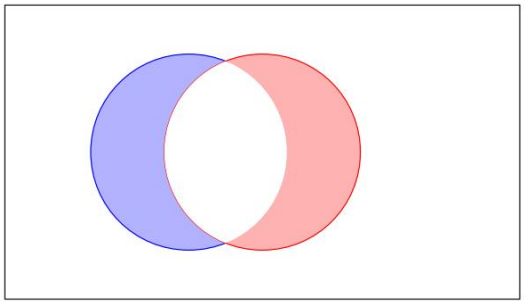


Figure: Venn diagram of $A \Delta B$.

4.10 algebra of sets

Commutative laws: $A \cup B = B \cup A$, $A \cap B = B \cap A$;

Associative laws: $A \cup (B \cup C) = (A \cup B) \cup C$, $A \cap (B \cap C) = (A \cap B) \cap C$;

Identity laws(恒等率): $A \cup \varnothing = A$, $A \cup U = U$, $A \cap U = A$, $A \cap \varnothing = \varnothing$;

Distributive laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;

Complement laws: $A \cup \sim A = U$, $\sim U = \varnothing$, $\sim (\sim A) = A$, $A \cap \sim A = \varnothing$, $\sim \varnothing = U$

De Morgan's laws: $\sim (A \cup B) = \sim A \cap \sim B$, $\sim (A \cap B) = \sim A \cup \sim B$

4.11 Reflection

The following statements hold:

- $\emptyset \in \{\emptyset\}$ but $\emptyset \notin \emptyset$;
- $\emptyset \subseteq \{5\}$;
- $\{2\} \nsubseteq \{\{2\}\}$ but $\{2\} \in \{\{2\}\}$;
- $\{3, \{3\}\} \neq \{3\}$.

4.12 Power Set

The power set $Pow(A)$ of a set A is the set of all subsets of A. In other words,

$Pow(A) = \{C \mid C \subseteq A\}$

Example:

Let $A = \{1, 2, 3\}$. Then

$Pow(A) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}$

Observation For all sets A and B: $Pow(A \cap B) = Pow(A) \cap Pow(B)$.

Proof We have to show for any set C that $C \in Pow(A \cap B)$ if and

only if $C \in Pow(A)$ and $C \in Pow(B)$. But

$C \in Pow(A \cap B) \Leftrightarrow C \subseteq A \cap B$

$\Leftrightarrow C \subseteq A$ and $C \subseteq B \Leftrightarrow C \in Pow(A)$ and $C \in Pow(B)$.

Observation There exist sets A and B such that $Pow(A \cup B) \mathrel{!=} Pow(A) \cup Pow(B)$.

4.13 Cardinality of sets

The cardinality of a *finite* set S is the number of elements in S, and is denoted by $|S|$. If A and B are sets then $|A \cup B| = |A| + |B| - |A \cap B|$.

$$|A \cup B \cup C| = |A|+|B|+|C| - |A \cap B| - |A \cap C| - |B \cap C|+|A \cap B \cap C|$$

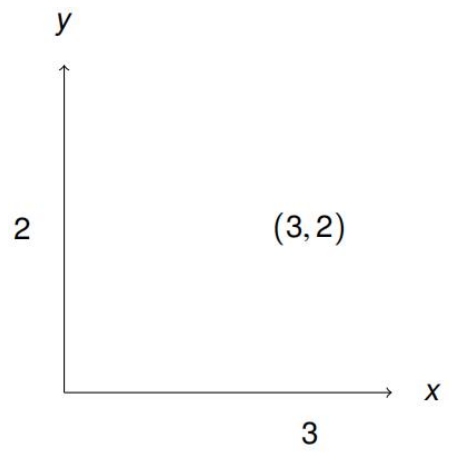
4.14 Ordered pairs

Definition The cartesian product $A \times B$ of sets A and B is the *set* consisting of all pairs (a, b) with $a \in A$ and $b \in B$, i.e., $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$.

Note that $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$

$\{1, 2\} = \{2, 1\}$ but $(1, 2) \mathrel{!=} (2, 1)$.

The set $R \times R$, or R_2 as it is often written, consists of all pairs of real numbers (x, y). R_2 is called the *Cartesian plane*



The cartesian product $A_1 \times A_2 \times \cdots \times A_n$ of sets $A_1, A_2, \ldots, A_n$ is defined by $A_1 \times A_2 \times \cdots \times A_n = \{(a_1, \ldots, a_n) \mid a_1 \in A_1, \ldots, a_n \in A_n\}$.

Here $(a_1, \ldots, a_n) = (b_1, \ldots, b_n)$ if and only if $a_i = b_i$ for all $1 \le i \le n$.

If $A_1, \ldots, A_n$ are all the same set A, then we write A^n to refer to the set $A_1 \times \cdots \times A_n$

4.15 Bit Strings of length n n 位位串

Let $B = \{0, 1\}$. B^n consists of all lists of zeros and ones of length n.

These are called *bit strings* of length n.

Bit strings can be used to represent the subsets of a set.

Suppose we have a set $S = \{s_1, \ldots, s_n\}$.

If we have a subset $A \subseteq S$, the characteristic vector of A is the n-bit string $(b_1, \ldots, b_n) \in B^n$ where

$$b_i = \begin{cases} 1 & \text{if} \quad s_i \in A \\ 0 & \text{if} \quad s_i \notin A \end{cases}$$

Let $S = \{1, 2, 3, 4, 5\}$, $A = \{1, 3, 5\}$ and $B = \{3, 4\}$.

- The characteristic vector of A is $(1, 0, 1, 0, 1)$.
- The characteristic vector of B is $(0, 0, 1, 1, 0)$.
- The characteristic vector of $A \cap B$ is $(0, 0, 1, 0, 0)$.
- The characteristic vector of $A \cup B$ is $(1, 0, 1, 1, 1)$.

5. Relations

5.1 Binary relation

Definition A binary relation between two sets A and B is a subset R of the Cartesian product $A \times B$. If $A = B$, then R is called a

binary relation on A.

We can form the set R of ordered pairs $(s, c) \in S \times C$ with the property that student s is registered for the module c. Then $R = \{(s, c) \in S \times C \mid s \text{ registered for } c\}$.

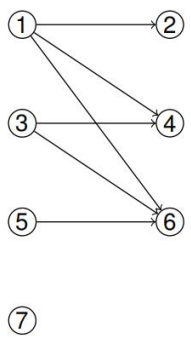
This subset of $S \times C$ captures the relationship *registered for*.

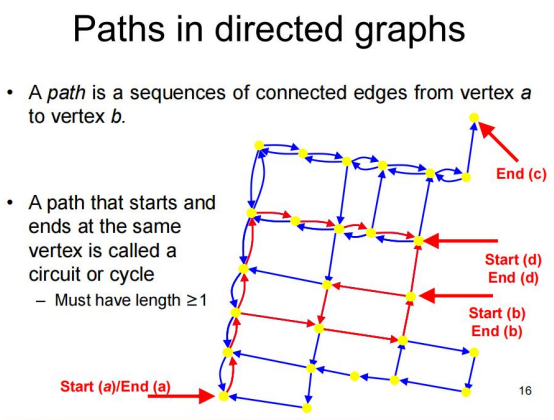
5.2 Directed graphs

(Representation of binary relations)

Let A and B be two finite sets and R a binary relation between these two sets (i.e., $R \subseteq A \times B$). We represent the **elements** of these two sets as **vertices** of a graph.For each $(a, b) \in R$, we draw an **arrow linking** the related elements. This is called the directed graph (or digraph) of R

Consider the relation V between $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6\}$ such that $V = \{(x, y) \in A \times B \mid x < y\}$.



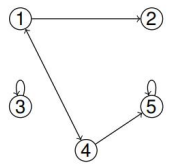


5.3Digraphs of binary relations on a single set

Recall that a binary relation between a set A and itself is called "a binary relation on A".

Consider the relation $V \subseteq A \times A$ where $A = \{1, 2, 3, 4, 5\}$ and

$$V = \{(1, 2), (3, 3), (5, 5), (1, 4), (4, 1), (4, 5)\}.$$



5.4 Matrices

(Representation of binary relations)

Let $A = \{a_1, \ldots, a_n\}$, $B = \{b_1, \ldots, b_m\}$ and $R \subseteq A \times B$.

We represent R by an array M of n rows and m columns.

The entry in row i and column j of this matrix is given by $M(i, j)$ where

$$M(i,j) = \begin{cases} T & \text{if} \quad (a_i, b_j) \in R \\ F & \text{if} \quad (a_i, b_j) \notin R \end{cases}$$

ex :Let $A = \{1, 3, 5, 7\}$, $B = \{2, 4, 6\}$, and

$U = \{(x, y) \in A \times B \mid x + y = 9\}$

Assume an enumeration $a_1 = 1$, $a_2 = 3$, $a_3 = 5$, $a_4 = 7$ and

$b_1 = 2$, $b_2 = 4$, $b_3 = 6$. Then M represents U, where

$M =$

F F F

F F T

F T F

T F F

5.5Unary Relations

Unary relations are just subsets of a set (本质上就是子集)

Example: The unary relation `EvenPositiveIntegers` on the set $\mathbb{Z}^+$ of positive integers is

$$\{x \in \mathbb{Z}^+ \mid x \text{ is even}\}.$$

5.6 Infix notation for binary relations

If R is a binary relation then we write xRy whenever $(x, y) \in R$.

The predicate xRy is read as x is R-related to y.

5.7 Properties of binary relations

A binary relation R on a set A is

reflexive when xRx for all $x \in A$.

$(\forall x)(x \in S \rightarrow (x,x) \in R)$

symmetric when xRy implies yRx for all $x, y \in A$;

$(\forall x)(\forall y)(x \in S \wedge y \in S \wedge (x,y) \in R \rightarrow (y,x) \in R)$

antisymmetric when xRy and yRx imply $x = y$ for all $x, y \in A$;

$(\forall x)(\forall y)(x \in S \wedge y \in S \wedge (x,y) \in R \wedge (y,x) \in R \rightarrow x=y)$

transitive when xRy and yRz imply xRz for all $x, y, z \in A$.

$(\forall x)(\forall y)(\forall z)(x \in S \wedge y \in S \wedge z \in S \wedge (x,y) \in R \wedge (y,z) \in R \rightarrow (x,z) \in R)$

状态	例子	解释
1. 既是对称又是反对称	$R = \{(1,1),(2,2)\}$	只有自环。没有 (a,b) 这种不同的配对，所以不违反任何规则。
2. 是对称，不是反对称	$R = \{(1,2),(2,1)\}$	有双向车道，所以是对称。违反了反对称。
3. 是反对称，不是对称	$R = \{(1,2)\}$	只有单向车道。违反了对称。符合反对称。
4. 既不对称，也不反对称	$R = \{(1,2),(2,1),(1,3)\}$	**这就是反例！** 因为有 $(1,2)(2,1)$，所以不是反对称。

In the directed graph representation

reflexive if there is always an arrow from every vertex to itself;

symmetric if whenever there is an arrow from *x* to *y* there is

also an arrow from *y* to *x*;

antisymmetric if whenever there is an arrow from *x* to *y* and

x , *y*, then there is no arrow from *y* to *x*;

transitive if whenever there is an arrow from *x* to *y* and from *y*

to *z* there is also an arrow from *x* to *z*.

5.8 Reflexive closure

The reflexive closure of R is $R \cup \Delta$

– Where $\Delta = \{ (a,a) \mid a \in R \}$

• Called the “diagonal relation”(对角关系)

5.9 Symmetric closure

In order to find the symmetric closure of a relation R, we add

an edge from a to b,<u>where there is already an edge from b to a</u>.

The symmetric closure of R is $R \cup R^{-1}$

– If $R = \{ (a,b) \mid \ldots \}$

– Then $R^{-1} = \{ (b,a) \mid \ldots \}$

5.10 Transitive Closure

Given a binary relation R on a set A, the transitive closure R∗ of R

is the (uniquely determined) relation on A with the following properties:

R ∗ is transitive;

R ⊆ R ∗ ; 完全包含原来的集合 满足上面要求的最小集合

If S is a transitive relation on A and R ⊆ S, then R ∗ ⊆ S (Rt is a subset of any other

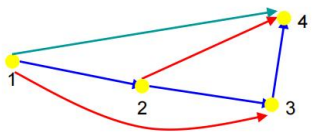
relation on S that includes R, and is transitive.)

$$Rt = R \cup \Delta$$

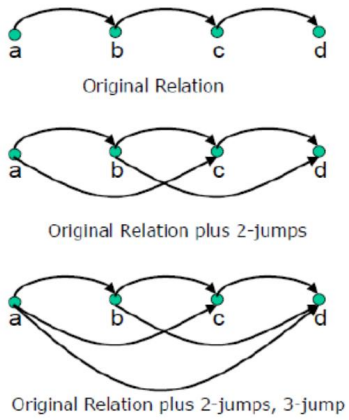
$$(a,b) \in Rt \wedge (b,c) \in Rt \rightarrow (a,c) \in Rt$$

Transitive closure

- We will study different algorithms for determining the transitive closure (in *S2*)
- red means added on the first repeat
- teal means added on the second repeat



Graphical Construction of Transitive Closure



5.11 Equivalence Relations

A binary relation R on a set A is called an *equivalence relation* if it is reflexive, transitive, and symmetric.

The *equivalence class* E_x of any $x \in A$ is defined by

$E_x = \{y \mid yRx\}$. 满足 yRx 的 x 的等价类

$E_{\frac{1}{2}} = \{\ldots, -2\frac{1}{2} - 1\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 1\frac{1}{2}, 2\frac{1}{2}, \ldots\}$ is the equivalence class of $\frac{1}{2}$.

这里可以说是 0.5 的等价类 也可以是 1.5 的等价类

5.12 Partial orderings

A partial ordering (or partial order) is a relation that is *reflexive*, *antisymmetric*, and *transitive*:

• Recall that antisymmetric means that if $(a,b) \in R$, then $(b,a) \notin R$ unless $b = a$.

• Thus, (a,a) is allowed to be in R.

• But since it is reflexive, all possible (a,a) must be in R.

A set S with a partial ordering R is called a *partially ordered set*, or *poset:* • Denoted by (S,R)

The symbol $\succ$ is used to represent *any* relation on the set S when discussing partial orders:

- Not just the less than or equals to relation.
- Can represent $\leq$, $\geq$, $\subseteq$, etc.
- Thus, $a \succ b$ denotes that $(a,b) \in \succ$.
- The poset is $(S,\succ)$.

Comparable: The elements a and b of a poset (S, ➢) are called ***comparable*** if either a ➢ b or b ➢ a.

2|4（2 能整除 4） 2|6 2，4 可比较 2，6 可比较

Incomparable: When a and b are elements of S such that neither a ➢ b nor b ➢ a, a and b are called ***incomparable***.

4，6 不可比较

Predecessor/Successor:前驱/后继

Let (**Z**,≤) be a partially ordered set. If x ≤ y but x≠y, we write x＜y and say that x is a **predecessor** of y or y is a **successor** of x.

Immediate predecessor(直接前驱)

A given y may have many predecessors, but if x＜y and there is no z with x＜z＜y, then x is an **immediate predecessor** of y.

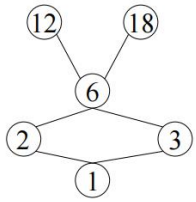
Consider the relation "x divides y" on {1,2,3,6,12,18}.

- Write the ordered pairs (x,y) of this relation.
  (1,1),(1,2),(1,3),(1,6),(1,12),(1,18),(2,2),(2,6),
  (2,12),(2,18),(3,3),(3,6),(3,12),(3,18),(6,6),(6,12),
  (6,18),(12,12),(18,18).

- Write all the predecessors of 6.
  1,2,3

- Write all the immediate predecessors of 6.
  2,3

Hasse Diagram



• **Visual** depiction of a partially ordered set.

• Each of the elements of S is represented by a dot, called a node, or vertex, of the diagram.

• If x is an ***immediate predecessor*** of y, then the node for y is placed **above** the node for x and the two nodes are connected by a **straight-line** segment.

• Two nodes in a Hasse diagram should never be joined by a horizontal line.(no 水平线)

5.13 Totally ordered

If $(S,\succ)$is a poset and **every two elements of *S* are *comparable***, *S* is called totally *ordered* or *linearly ordered* set, and $\succ$ is called a ***total order*** or a ***linear order***.A totally ordered set is also called a Chain 全序是偏序的一种特殊情况

5.14 Partition of a set

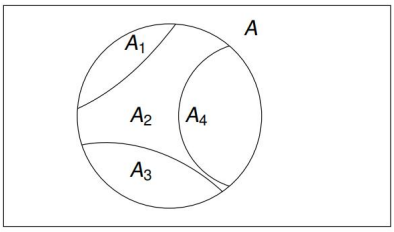
A *partition* of a set A is a collection of non-empty subsets

$A_1, \ldots, A_n$ of A satisfying:

$A = A_1 \cup A_2 \cup \cdots \cup A_n$;

$A_i \cap A_j = \varnothing$ for i, j.

The A_i are called the blocks of the partition.



5.15 Inverse Relation

Given a relation $R \subseteq A \times B$, we define the *inverse relation* $R^{-1} \subseteq B \times A$ by

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}.$$

5.16 Composition of Relations

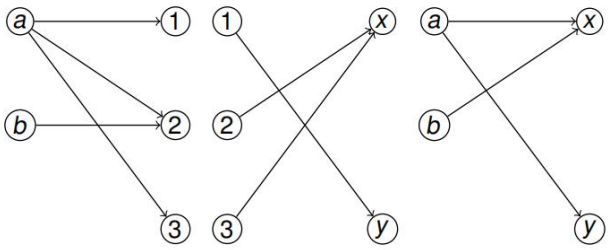
Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The composition of R and S, denoted by $S \circ R$, is the binary relation between A and C given by

$$S \circ R = \{(a, c) \mid \text{exists } b \in B \text{ such that } aRb \text{ and } bSc\}.$$

Example: If R is the relation *is a sister of* and S is the relation *is a parent of*, then

$S \circ R$ is the relation *is an aunt of*,

$S \circ S$ is the relation *is a grandparent of*.



$$R: \begin{bmatrix} T & T & T \\ F & T & F \end{bmatrix} \quad S: \begin{bmatrix} F & T \\ T & F \\ T & F \end{bmatrix} \quad S \circ R: \begin{bmatrix} T & T \\ T & F \end{bmatrix}$$

乘法是 AND 逻辑 加法是 OR 逻辑

5.17 Matrix representation of compositions

Then the entries $P(i, j)$ of the logical matrix P representing $S \circ R$ are given by

$P(i, j) = T$ if there exists l with $1 \leq l \leq m$ such that $M(i, l) = T$

and $N(l, j) = T$.

$P(i, j) = F$, otherwise.

We write $P = MN$.

Transitivity and Composition

We saw earlier that a relation S is transitive if and only if $S \circ S \subseteq S$.

This is because $S \circ S = \{(a, c) \mid \text{exists } b \text{ such that } aSb \text{ and } bSc\}$.

Let S be a relation. Set $S^1 = S$, $S^2 = S \circ S$, $S^3 = S \circ S \circ S$, and so on.

$$\begin{bmatrix} T & F & F & T & F \\ F & T & F & F & T \\ F & F & T & F & F \\ T & F & T & F & F \\ F & T & F & T & F \end{bmatrix} \begin{bmatrix} T & F & F & T & F \\ F & T & F & F & T \\ F & F & T & F & F \\ T & F & T & F & F \\ F & T & F & T & F \end{bmatrix} = \begin{bmatrix} T & F & \color{red}{T} & T & F \\ F & T & F & T & T \\ F & F & T & F & F \\ T & F & T & T & F \\ T & T & T & F & T \end{bmatrix}$$

$$R \circ R = \{(a, c) \mid \text{ exists } b \in A \text{ such that } aRb \text{ and } bRc\}.$$

Note (in red) that there are pairs (a, c) that are in $R \circ R$ but not in R. Hence, R is not transitive.

6. Functions

6.1 Cartesian product

$$A \times B = \{x, y \mid x \in A \text{ and } y \in B\}$$

Example: Find the cartesian products of set A ={1,2,3} and B={3,4,5}.

Solution:

Following the above definition, let Cartesian product be X,

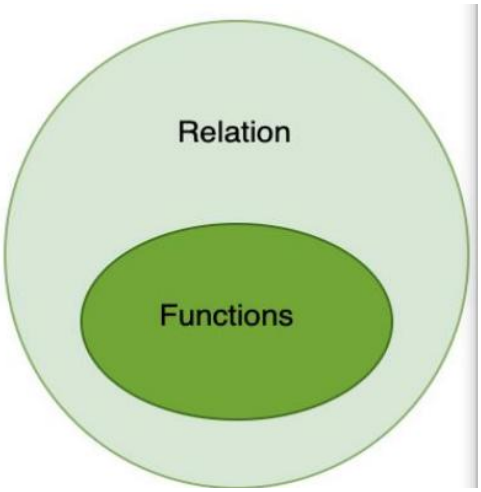
X = A x B
= {(1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5)}

6.2 A function is a special kind of relation

Relation

A relation from set A to set B is a subset of the cartesian product set A x B.

The subset is made up by describing a relationship between the first element and the second element of elements in A x B.



6.3Definition

(1)a 属于 A there is precisely one pair of the form (a,b) in R

A function from a set ***A*** to a set ***B*** is a binary relation ***R*** in which *every element* of ***A*** is associated with a *uniquely specified* element of ***B.***

(2) mapping or transformation

A function ***f*** from **A** to **B**, written ***f : A → B***, is a rule that assigns to each input value a∈A *exactly one* output value b∈B; the unique value **b** assigned to **a** is denoted by ***f (a)***.

6.4 Domain codomain and range

Suppose $f: A \to B$.

- A is called the ***domain*** of f.
- B is called the ***codomain*** of f.
- The ***image* of *x* under *f*** is the value $f(x)$;
- if R is any subset of A, then $f(R) = \{f(r) \mid r \in R\}$ is **the set of images of elements of *R*** and is called **the *image of R***.
- The ***image*** *or* ***range*** ***f(A)* of *f*** is $f(A) = \{f(x) \mid x \in A\}$

R 是 A 的一个子集，f(R)只看这一个子集 R 映射出来的函数值

(r 属于 R)　f(A)是对 A 中所有的元素 x 的映射

Codomain 是大的>= domain 是小的

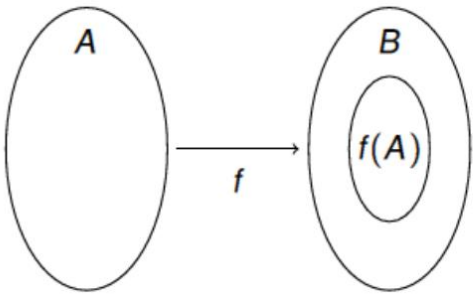


Figure: The image values of f is called **image** or **range** of f

Image of *f* = { b ∈ B: there exists a ∈ A such that f(a) = b}
f(A) ⊆ B – Image of f is a subset of B

image of *A* under *f* = range of *f*

6.5 All functions are relations, but not all relations are functions

6.6 Injective(one-to-one) functions

Definition. Let $f : A \to B$ be a function. We call f an ***injective*** (or **one-to-one**) function if

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2 \text{ for all } a_1, a_2 \in A.$$

Injective functions never repeat values,

different inputs give different outputs

Proof example:

If the function f: $\mathbf{R} \to \mathbf{R}$ is defined by the rule $f(x) = 4x - 1$, for each real number x, then f is one-to-one.

Proof: Suppose x_1 and x_2 are real numbers such that $f(x_1) = f(x_2)$. *[We must show that $x_1 = x_2$.]* By definition of f,

$$4x_1 - 1 = 4x_2 - 1.$$

Adding 1 to both sides gives

$$4x_1 = 4x_2,$$

and dividing both sides by 4 gives

$$x_1 = x_2,$$

[as was to be shown].

6.7 Surjective functions: codomain=range

Definition. $f : A \to B$ is **surjective** (or onto) if the range of f coincides with the codomain of f.

This means that for every $b \in B$ there exists $a \in A$ with $b = f(a)$.

代入 y=f(x)反解出 x / x 用 y 表示 带入 f(x)中

If f: $\mathbf{R} \to \mathbf{R}$ is the function defined by the rule $f(x) = 4x - 1$ for each real number x, then f is onto.

Proof: Let $y \in \mathbf{R}$. *[We must show that $\exists x$ in $\mathbf{R}$ such that $f(x) = y$.]* Let $x = (y + 1)/4$. Then x is a real number since sums and quotients (other than by 0) of real numbers are real numbers. It follows that

$$\begin{aligned} f(x) &= f\left(\frac{y+1}{4}\right) && \text{by substitution} \\ &= 4 \cdot \left(\frac{y+1}{4}\right) - 1 && \text{by definition of } f \\ &= (y+1) - 1 = y && \text{by basic algebra,} \end{aligned}$$

[as was to be shown].

6.8 Bijective（双射）

We call f bijective if f is both injective and surjective

6.9 Inverse Functions

Since any function $f: A \to B$ is a relation, we can form the inverse relation f^{-1}.

If this inverse relation is itself a function, we say that f is an *invertible* function and write

$f^{-1}: B \to A$

for the *inverse* function.

If $f(a) = b$ then (a, b) is a pair in the function, so (b, a) is a pair in the inverse, so $f^{-1}(b) = a$.

6.10 A function f is invertible if and only if it is a bijection

6.11 Composition of functions

If $f: A \to$ B and g : $B \to$ C are functions, then the composition relation $g \circ f$ between A and C consists of all pairs (a, c) where,

for some $b \in B$, (a, b) $\in f$ and (b, c) $\in$ g, $g \circ f$ is a function as well.

It is given by

$$(g \circ f)(x) = g(f(x)).$$

6.12 invertible function of the composition functions

Let $f : A \to B$.

A function $g : B \to A$ is the inverse of f if

$$f \circ g = 1_B \text{ and } g \circ f = 1_A.$$

identity on B　　identity on A

$1_B(y) = y$（对于所有 B 中的元素 y）恒等函数

6.13 The pigeonhole principle PHP

Let $f : A \to B$ be a function where A and B are finite sets.

Suppose that A contains n elements labelled $a_1, \ldots, a_n$.

The *pigeonhole principle* states that if $|A|>|B|$ then at least one value of f occurs more than once.

In other words, we have $f(a_i) = f(a_j)$ for some i and j where $i \neq j$.

So, there can be **no one-to-one function** $f : A \to B$.

7.Propositional Logic

7.1 Definition

A ***proposition*** is a *declarative sentence* that is either *true* or *false*.

A ***proposition*** is a *possible condition* of the world that is either *true* or *false*

The area of logic that deals with propositions is called propositional logic. Propositional Logic is concerned with propositions and their interrelationships.(关注命题本身以及之间的联系)

Propositional Variables

A variable that represents propositions is called a *propositional variable.* Propositional variables are called atomic formulas.

p0,pi,p2...

Formulas (wellformed formulas (wffs)

can be built from the proposition letters using the propositional connectives and parentheses.

Syntactically(语法上) Semantics(语义上的)

"Parsing" （解析）the formula means determining how the formula is built, following the rules.
We are going to present the propositional logic as a formal language:
- we first present the syntax of the language
- then the semantics of the language.

7.2 Syntax of Propositional Logic

The *alphabet of propositional logic* consists of

- an infinite set $p, q, r, p_o, p_i, p_2, \ldots$ of *atomic formulas* (or propositional variables);
- the logical connectives $\neg$ ("not"), $\wedge$ ("and"), and $\vee$ ("or");
- brackets: (and).

7.3 Formulas of propositional logic

The set P of all *formulas of propositional logic* is defined inductively:

- all atomic formulas are formulas;
- if *P* is a formula, then ¬P is a formula;
- if P and *Q* are formulas, then (P ∧ *Q*) is a formula;
- if P and *Q* are formulas, then (P v *Q*) is a formula;
- Nothing else is a formula.

7.4

formula corresponds to a unique binary tree.

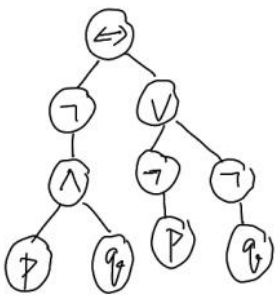
We can evaluate the formula by:

- giving each propositional variable an interpretation.

- defining the meaning of each logical connective

- propagate the truth values from the leaves to the root in a unique way, so that we get an unambiguous evaluation of each formula.

7.5 Parse Tree



$\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$

- The parse tree of an atomic formula has one node, labelled with the formula.
- The parse tree of a formula ¬*P* has a root labelled "¬". The child of the root is the parse tree for the formula *P*.
- The parse tree of a formula (*P* ∧ *Q*) has a root labelled "∧". The left child of the root is the parse tree for the formula *P*. The right child of the root is the parse tree for the formula *Q*.
- The parse tree of a formula (*P* ∨ *Q*) has a root labelled "∨". The left child of the root is the parse tree for the formula *P*. The right child of the root is the parse tree for the formula *Q*.

7.6 Truth Values 解释（规定）

An *interpretation* I is a function which assigns to any atomic formula p_i a truth value $I(p_i) \in \{0, 1\}$.

If $I(p_i) = 1$, then p_i is called *true* under the interpretation I.

If $I(p_i) = 0$, then p_i is called *false* under the interpretation I.

Given an assignment I we can compute the truth value of compound formulas step by step using so-called *truth tables*

Negation

The negation $\neg P$ of a formula P is true when P is false and false otherwise:

Definition Suppose an interpretation I is given and we know the value $I(P)$. Then the value $I(\neg P)$ is computed by

$$I(\neg P) = \begin{cases} 0 & \text{if} \quad I(P) = 1 \\ 1 & \text{if} \quad I(P) = 0 \end{cases}$$

Corresponding truth table:

P	$\neg P$
1	0
0	1

Conjunction

The conjunction $(P \wedge Q)$ is true if and only if both P and Q are true.

Definition Suppose an interpretation I is given and we know $I(P)$ and $I(Q)$. Then

$$I(P \wedge Q) = \begin{cases} 1 & \text{if} \quad I(P) = 1 \text{ and } I(Q) = 1 \\ 0 & \text{if} \quad I(P) = 0 \text{ or } I(Q) = 0 \end{cases}$$

Corresponding truth table:

P	Q	$(P \wedge Q)$
1	1	1
1	0	0
0	1	0
0	0	0

Disjunction

The disjunction $(P \vee Q)$ is true if and only P is true or Q is true.

Definition Suppose an interpretation I is given and we know $I(P)$ and $I(Q)$. Then

$$I(P \vee Q) = \begin{cases} 1 & \text{if} \quad I(P) = 1 \text{ or } I(Q) = 1 \\ 0 & \text{if} \quad I(P) = 0 \text{ and } I(Q) = 0 \end{cases}$$

Corresponding truth table:

P	Q	$(P \vee Q)$
1	1	1
1	0	1
0	1	1
0	0	0

True under an interpretation

we can compute the truth value of any formula P under I.

If $I(P) = 1$, then P is called true under the interpretation I.

If $I(P) = 0$, then P is called false under the interpretation I.

$p \to q$ implication

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

$p \leftrightarrow q$

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

List the Interpretations I such that $P = ((p_1 \vee \neg p_2) \wedge p_3)$ is true under I. $I(p_1 \vee \neg p_2) = 1$ $I(p_3) = 1$

	p_1	p_2	p_3
①	1	0	1
②	1	1	1
③	0	0	1

Truth table for $(\neg P \vee Q)$

P	Q	$\neg P$	$(\neg P \vee Q)$
1	1	0	1
1	0	0	0
0	1	1	1
0	0	1	1

$(\neg P \vee Q)$ represents the assertion 'if P is true, then Q is true'.
Define a 'new' connective $\to$ by:

$$(P \to Q) \equiv (\neg P \vee Q).$$

implication

$P \to Q$ is true always except when P is true and Q is false

当 p=0 时，无法推出 Q 是否为 1 所以 Q=0/1 都对

Truth table for $((P \to Q) \wedge (Q \to P))$

P	Q	$(P \to Q)$	$(Q \to P)$	$((P \to Q) \wedge (Q \to P))$
1	1	1	1	1
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

$((P \to Q) \wedge (Q \to P))$ represents the assertion 'P is true if and only if Q is true'. Define a 'new' connective $\leftrightarrow$ by:

$$(P \leftrightarrow Q) := ((P \to Q) \wedge (Q \to P)).$$

biconditional

$P \leftrightarrow Q$ is true when P and Q have the same truth value, i.e., they are both true or both false.

7.7Tautology(重言式：永远为真的公式)

A tautology is a formula which is true under all interpretations

Example All formulas of the form $(P \vee \neg P)$ are tautologies, because

$$I(P \vee \neg P) = 1$$

for all interpretations I:

P	$\neg P$	$(P \vee \neg P)$
1	0	1
0	1	1

Define a 'new' connective $\top$ by $\boxed{\top := (p_1 \vee \neg p_1).}$

7.8Contradiction（矛盾式：永远为假）

A contradiction is a formula which is false under all interpretations.

Example All formulas of the form $(P \wedge \neg P)$ are contradictions, because

$$I(P \wedge \neg P) = 0$$

for all interpretations I:

¬⊤ = ⊥（“永远真”的否定是“永远假”），
¬⊥ = ⊤（“永远假”的否定是“永远真”）。

P	$\neg P$	$(P \wedge \neg P)$
1	0	0
0	1	0

Define a ‘new’ connective $\bot$ by $\boxed{\bot := (p_1 \wedge \neg p_1).}$

假

7.9Semantic consequence (语义后承，逻辑推导有效性)

Definition Suppose Γ is a finite set of formulas and P is a formula. Then P follows from Γ (“is a semantic consequence of Γ”) if the following implication holds for every interpretation I:

$$\text{If } I(Q) = 1 \text{ for all } Q \in \Gamma, \text{ then } I(P) = 1.$$

This is denoted by

$$\Gamma \models P.$$

设 Γ 是一组公式（前提），P 是一个公式（结论）。如果所有使 Γ 中所有公式都为真的解释，都能使 P 为真，就说 P 是 Γ 的语义后承，记为 Γ ⊨ P（读作“Γ 语义蕴含 P”）。

核心逻辑：

“前提都真时，结论一定真”——推导是“有效的”，和前提本身是否为真无关，只看“前提真→结论真”的必然性。

前提：所有猫会飞，我家宠物是猫 结论：我家宠物会飞

翻译成人话就是： 我们在检查逻辑严不严谨时，只看一种情况——**当前提都为真的时候**。

- 我们**不关心**前提为假的时候发生了什么（那是废话）。
- 我们**只关心**：在所有让“证据包” Γ 成立的世界里（$I(Q) = 1$），结论 P 是不是也**被迫**成立（$I(P) = 1$）？
- 如果是，那就叫 $\Gamma \models P$（推理有效）。
- 如果有哪怕一次例外（前提都对，结论却错了），那就推导不出来。

7.10 Logical equivalence

Definition Two formulas P and Q are called equivalent if they have the same truth value under every possible interpretation. In other words, P and Q are equivalent if $I(P) = I(Q)$ for every interpretation I. This is denoted by

$$P \equiv Q.$$

- 等价是“双向的”：P ≡ Q 等价于 {P} ⊨ Q 且 {Q} ⊨ P;

- 语义后承是“单向的”：Γ ⊨ P 不一定意味着 P ⊨ Γ。

Theorem The relation $\equiv$ is an equivalence relation on $\mathcal{P}$.

Proof

- $\equiv$ is reflexive, since $(P \leftrightarrow P)$ is a tautology
- $\equiv$ is transitive, since $P \equiv Q$ and $Q \equiv R$ implies $P \equiv R$.
- $\equiv$ is symmetric, since $P \equiv Q$ implies $Q \equiv P$.

7.11 Laws for equivalence

The following equivalences can be checked by truth tables:

- Associative laws: 结合律

  $(P \vee (Q \vee R)) \equiv ((P \vee Q) \vee R)$, $(P \wedge (Q \wedge R)) \equiv ((P \wedge Q) \wedge R)$;

- Commutative laws: 交换律

  $(P \vee Q) \equiv (Q \vee P)$, $(P \wedge Q) \equiv (Q \wedge P)$;

- Identity laws: 同一律 (和真/假结合)

  $(P \vee \bot) \equiv P$, $(P \vee \top) \equiv \top$, $(P \wedge \top) \equiv P$, $(P \wedge \bot) \equiv \bot$;

- Distributive laws: 分配律

$$(P\wedge(Q\vee R)) \equiv ((P\wedge Q)\vee(P\wedge R)),\ (P\vee(Q\wedge R)) \equiv ((P\vee Q)\wedge(P\vee R));$$

- Complement laws: 补余律

$$P \vee \neg P \equiv \top,\ \neg\top \equiv \bot,\ \neg\neg P \equiv P, P \wedge \neg P \equiv \bot,\ \neg\bot \equiv \top;$$

- De Morgan's laws: 德摩根定律

$$\neg(P \vee Q) \equiv (\neg P \wedge \neg Q),\ \neg(P \wedge Q) \equiv (\neg P \vee \neg Q).$$

8.Predicate Logic

The Statement in Predicate Logic

The original statement can now be written symbolically using predicates, variables, and quantifiers:

$$\forall x\Big(Student(x) \rightarrow \exists y\,\big(Younger(x,y) \wedge Instructor(y)\big)\Big)$$

- This is a typical formula in predicate logic!
- Translating the formula back yields:

  "For every individual x, if x is a student, then there is an individual y such that x is younger than y and y is an instructor."

Translation into Predicate Logic II

"Not all birds can fly."

1. Pick suitable predicates: *Bird* (being a bird), *CanFly* (can fly)
2. Encode the sentence in predicate logic:

$$\neg\forall x\big(Bird(x) \rightarrow CanFly(x)\big)$$

"It is not the case that for all individuals x, if x is a bird, then x can fly."

Equivalently:

$$\exists x\big(Bird(x) \wedge \neg CanFly(x)\big)$$

"There is an individual that is a bird and that cannot fly."

This equivalence can be made precise (later lecture).

Equality and Function Symbols

Predicate logic has two additional features:

等词

1. Equality: allows to express that individuals are equal or not

Example: *"At least two students are registered for COMP118."*

$$\exists x\,\exists y\,\big(Registered(x, comp118) \wedge Registered(y, comp118) \wedge \neg x = y\big)$$

描述对象的映射

和谓词不同，函数符号的结果是"另一个对象"，不是真假：

2. Function symbols: allows to express functional dependencies between individuals

Example: *"Alice and Bob have the same mother."*

$$Mother(alice) = Mother(bob)$$

- 谓词：IsMotherOf(x, alice)（x是爱丽丝的妈妈，结果是真/假）；

- 函数：Mother(alice)（爱丽丝的妈妈，结果是"爱丽丝妈妈"这个对象）。

Predicate logic vs Function

Predicate is to show a comparison or a relation between two objects, such as:
Ex: IsMotherOf(x, alice) ; President(Obama, America)

Functions are to specify what a particular object is, such as:
Ex: Mother(alice) ; Human(Einstein)

A predicate is a function that returns true or false (Boolean function).

A function associates individuals and predicates their meanings.

Domain

论域（谈论的所有对象的集合

Domains (Domains of discourse): a *non-empty set of objects* (humans, animals, account numbers, …).

- In *mathematics*, the domain usually consists of numbers (*integers, rationals, reals, etc.*), vectors, geometric figures (*points, lines, triangles, spheres, etc.*), sets, graphs, etc.
- A domain in a *nonmathematical* discourse may consist of material objects, human beings, ideas or anything else.

Constants

常元

Constants: *concrete objects in the domain*. Some objects in the domain can be distinguished in a way that would allow us to make direct references to them, by giving them *names*.

e.g. in the domain of real numbers are 0, 1, π,e, etc.

Variables

变元

Variables: when we deal with *unknown or unspecified objects* from the domain of discourse.

- in natural language we usually use pronouns or other syntactic constructions

  Example "*If a man owes money to another man then that man hates the other man*"

- we use variables as *placeholders for the arguments of the various predicates and functions* we deal with.
- for variables we will be using *letters u, v, w, x, y, z,* possibly indexed.

  Example

  - the (unary) functions: f (x) = x + 1 and m(z) = "*the mother of z*",
  - the (binary) predicates x < y and "x loves y", etc.

Predicates

Predicates. A predicate takes object(s) in the domain as argument(s) and returns *true* or *false*.

They describe *properties* of objects or *relationships* between objects.

Example talking about integers - properties like being "*positive*", "*divisible by 3*", "*not greater than 2000*", etc.

- ***unary*** predicates 一元谓词：描述属性
- ***binary*** predicates - relating two objects: "*_is less than _*", "*_ is a son of _*", etc.
- ***n-ary*** predicates, relating *n* objects.

As long as we specify the meaning (*semantics*) of a predicate and the *objects* that it relates, that predicate becomes true or false.

语义解释（赋值真假）
公式本身没有真假，只有指定"论域+签名的解释"后，才会有真假。

We can connect predicates, by using propositional connectives

Function Symbol

Function symbols. We use functions to represent *operations* that, applied to one or several objects, *determine an object*.

Depending on the number of arguments, we talk *about unary, binary, etc., functions.*

> ***Example***
> *"the mother of_", "the older grandfather of_", etc.,* in the domain of humans.

Logical symbols, including:

(a) **propositional connectives**,
(b) **quantifiers** - to quantify over objects of our discourse,
量词

Auxiliary symbols, such as (,)
辅助

Signatures : predicate symbols, constant symbols, function symbols

A signature defines the "vocabulary" relative to which formulae can be expressed.

It is the set of all the symbols that can be used in a formula:

- Predicate symbols: *Student*, *Younger*, *Bird*, …
- 常元符号 Constant symbols: *alice*, *comp118*, *42*, …
- Function symbols: *Mother*, *Plus*, *Minus*, …

It also fixes an arity (number of parameters) for each predicate and function symbol in it.

$$S_K = \{Male, Female, Parent, Sibling, alice, Mother, Father\},$$

where

- *Male* and *Female* are predicate symbols of arity 1
- *Parent* and *Sibling* are predicate symbols of arity 2
- *alice* is a constant symbol,
- *Mother* and *Father* are function symbols of arity 1

$S_A = \{Smaller, 0, 1, Plus, Times\}$

- x is an even number:

$$\exists y\, x = Plus(y, y)$$

- x is prime:

$$Smaller(1, x) \wedge \forall y \forall z \left(Times(y, z) = x \rightarrow (y = 1 \vee y = x) \right)$$

Example: Students and Instructors

$$\forall x \Big(Student(x) \rightarrow \exists y \left(Younger(x, y) \wedge Instructor(y) \right) \Big)$$

- A possible domain: $D = \{\text{Alice}, \text{Bob}, \text{Carol}\}$
- To say that Alice and Bob are students, we could interpret the predicate symbol *Student* with the set:

$$\{\text{Alice}, \text{Bob}\}$$

- To say that Alice is younger than Bob, and Bob is younger than Carol, we could interpret *Younger* with the set:

$$\{(\text{Alice}, \text{Bob}),\ (\text{Bob}, \text{Carol}),\ (\text{Alice}, \text{Carol})\}$$

Proposition Logic vs Predicate Logic

A predicate is a sentence that contains one or more variables

A predicate is neither true nor false

A predicate becomes a proposition when the variables are substituted with specific values

The domain of a predicate variable is the **set of all values that may be substituted for the variable**

9.COMBINATORICS

9.1 product rule (Multiplication Principle)

If there are $n(A)$ *ways to do* A and $n(B)$ *ways to do* B, then the number of ways to do A and B is $n(A) \times n(B)$. This is true if the number of ways of doing A and B are independent;

$$|A_1 \text{ x } A_2 \text{ x } \bullet\bullet\bullet \text{ x } A_m| = |A_1| \bullet |A_2| \bullet \ldots \bullet |A_m|.$$

9.2 Sum Rule (addition principle)

$|A \cup B| = |A| + |B|$ as long as A and B are disjoint sets.

$$|A_1 \cup A_2 \cup \ldots \cup A_m| = |A_1| + |A_2| + \ldots + |A_m|$$

when A_i *jiao* $A_j = \varnothing$ for all i, j.

9.3 Subtraction Rule(Subtraction Principle)

the *principle of inclusion-exclusion:*

$$|A \cup B| = |A| + |B| - |A \cap B|$$

9.4 Division Rule (Division Principle)

If a task can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w, then there n/d ways to do the task.

9.5 Factorial Function

$N! = n(n-1)(n-2)\cdots 2\cdot 1$

9.6 Permutations

Formula for **permutations without repetition**

$$P(n,r) = \frac{n!}{(n-r)!}$$

Formula for **permutations with repetition**

$$P(n,r) = n^r$$

Permutations of Identical Objects

The number of permutations of n objects with n_1 identical objects of type 1, n_2 identical objects of type 2, …, and n_k identical objects of type k is:

$$\frac{n!}{n_1!\,n_2!\ldots n_k!}$$

In general, the number of permutations of n objects with r of the objects identical objects is:

$$\frac{n!}{r!}$$

Example How many words (including nonsense words) can be made from rearrangements of the word ALPACA?

$\frac{6!}{3!} = \frac{720}{6} = 120$. There are 6 letters in ALPACA and one of them, 'A' is repeated 3 times.

Example How many words can be made from rearrangements of the word BANANA?

$\{B, A, N, A, N, A\} = \{A, B, N\}$.
The 'A' is repeated 3 times.
The 'N' is repeated 2 times.
The 'B' is repeated once.
Hence the answer is $\frac{6!}{1!\cdot 2!\cdot 3!} = 60$.

9.7 Combinations

Without repetition

$$C(n,r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

$0 \le r \le n$

With Repetition(插板法)

$$C(n+r-1,r) = \binom{n+r-1}{r} = \frac{(n+r-1)!}{r!\,(n-1)!}$$

COMBINATIONS

If you have **n** objects to choose from

And you have **k** slots to put them in

The order doesn't matter
(each slot is the same)

All the prizes or slots or positions are the same.
i.e., every winner gets the same award

Without repetition

$$\frac{n!}{k!\,(n-k)!}$$

With repetition

$$\frac{(n+k-1)!}{k!\,(n-1)!}$$

PERMUTATIONS

If you have **n** objects to choose from

And you have **k** slots to put them in

The order does matter
(each slot is different)

All the prizes or slots or positions are different.
i.e., first place, second place etc.

Without repetition

$$\frac{n!}{(n-k)!}$$

With repetition

$$n^k$$

10.Probability Theory

10.1 Foundations

Experiment (Random process): Any action with unpredictable outcomes

Sample Space: is the set, Ω, of all possible outcomes of a random process or experiment

Event: A specific outcome. A ***subset*** of the sample space. {H, T} H, T, HT, $\emptyset$

Special Events

The *Null Event*, The empty event - ϕ $\qquad$ $\phi = \{\ \}$ = there are no outcomes

The *Entire Event*, The Sample Space - Ω

The empty event, ϕ, never occurs.
The entire event, Ω, always occurs.

The probability can be expressed as a *fraction*, a *decimal*, or a *percent*

If the probability of an event is **0**, the event is **impossible**.

If the probability of an event is **1**, the event is **certain**.

X' is the **complement** of an event X.
The probability that any event X does not occur is P(X'): $\boldsymbol{P(X') = 1 - P(X)}$

An **event** is a *subset* $X \subseteq \Omega$. The probability of the event is given by

$$P(X) = \sum_{x \in X} P(x). \qquad P(\emptyset) = 0 \text{ and } P(\Omega) = 1$$

Probability P(X): The probability of outcome X occurring.

$$P(X) = \frac{n(X)}{n(\Omega)} = \frac{n(X)}{N} = \frac{\text{number of outcomes in } X}{\text{total number of outcomes}}$$

Probability of two or more Independent Events

$$P(A \textit{ and } B) = P(A) \times P(B)$$

Mutually Exclusive Events 互斥事件

If the two events have nothing in common, then they are called *mutually exclusive events*, mutually exclusive events are similar to mutually exclusive sets.

Formally two events A and B are called mutually exclusive if both of them cannot occur simultaneously.

Events that **can't happen at the same time**.

If two events A and B are **disjoint** (never occur at the same time), then the probability of either event is *the sum of the probabilities of the two events*:

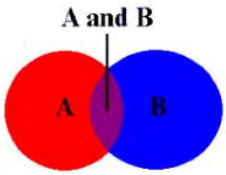
$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B).$$

Note: a union (∪) of two events occurring means that A or B occurs.

Note: disjoint events does not necessarily mean that they are independent.

Mutually Inclusive Events

Events have one or more outcomes in common.



If two events A and B are **not disjoint** (have some overlap with each other / can happen at the same time / occur in a single trial), then the probability of their union (the event that A or B occurs) is equal to *the sum of their probabilities minus the probability of their intersection.*

$$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$$

Conditional Probability

The probability of occurrence of any event X when another event Y in relation to X has already occurred is known as conditional probability.

$$P(X \mid Y)$$

The probability that this **WILL** happen

Assuming this already **HAS** happened

$$P(X \mid Y) = \frac{\frac{N(X \cap Y)}{N}}{\frac{N(Y)}{N}} = \frac{P(X \cap Y)}{P(Y)}$$

$N(X \cap Y)$ is the number of elements common to both X and Y. $N(Y)$ is the number of elements in Y, and it cannot be equal to zero. Let N represent the total number of elements in the sample space.

Probability without replacement

involves dependent events where the preceding event has an effect on the probability of the next event. The **sample space changes** for different events and the occurrence of the next event depends upon what happens in the preceding event.

$$P(B \mid A) = P(2^{nd}\ card\ is\ a\ Queen \mid 1^{st}\ card\ is\ a\ King) = \frac{4}{51} \approx 0.078$$

Probability with replacement

What is the probability of drawing twice from a deck and getting the **king** of diamonds **then** putting the card back in, then drawing the **queen** of diamonds?

THIS IS CALLED **REPLACEMENT**

The events are conditional $P(X) \times P(Y|X)$

$$\frac{1}{52} \times \frac{1}{52} = \frac{1}{2704}$$

Random Variable (function)

For each element of an experiment's sample space Ω, the random variable can take on exactly one value.

X = {0, 1}

Note: We could choose Heads=100 and Tails=150 or other values if we want! It is our choice.

Expected Value

The **expected value** in general, the value that is most likely the result of the next repeated trial of a statistical experiment.

Expected value uses all possible outcomes and their probabilities of occurring to find the weighted average of the data in the data set.

$$E(f) = \sum_{x \in \Omega} f(x)P(x)$$

Linearity of expectation

If f and g are two random variables, then

$$E[f+g] = E[f] + E[g].$$

More linearity of expectation

Suppose f is a random variable. Define the random variable g as follows: For every $x \in \Omega$, $g(x) = cf(x)$. Then $E[g] = cE[f]$.

Not linearity of expectation

If f and g are two random variables, then some students assume that $E[f \times g] = E[f] \times E[g]$. This is not always true.

只有相互独立时，第三条这个才会成立